Robust and Explainable AQI Forecasting under Missing Pollutant: An Imputation-Based Machine Learning Approach

Abstract— Machine-learning-based AQI prediction can achieve very low error when pollutant concentrations are available, but such performance may be sensitive to missing observations in the same pollutant variables used by the model. This study evaluates the robustness of an XGBoost AQI prediction model under controlled missingness of PM2.5, PM10, NO2, SO2, CO, and O3. Measurements from the NSIT Dwarka monitoring station in Delhi are used, with chronological training on 2020–2024 and testing on the unseen 2025 observations. Missingness levels of 5%, 10%, 20%, and 30% are imposed on pollutant inputs and six imputation strategies—Mean, Median, Forward Fill, Linear Interpolation, KNN, and XGBoost-based imputation—are compared. The complete-data baseline achieves MAE 0.7774, RMSE 1.2228, and R^2 0.9999. Under 30% missingness, Mean imputation gives MAE 6.1867 and R^2 0.9950, while Forward Fill gives MAE 10.1538 and R^2 0.9902. Across the four missingness levels, Mean and Median imputation maintain the lowest average MAE, whereas Forward Fill shows the largest degradation. SHAP analysis further shows that PM2.5 remains the dominant predictor and PM10 the second most influential feature across the tested missingness levels. The results demonstrate that predictive accuracy decreases progressively as pollutant observations become unavailable, while the dominant feature structure remains comparatively stable.

Keywords— AQI, XGBoost, missing data, imputation, pollutant dependency, SHAP, PM2.5, PM10

I. INTRODUCTION

Air Quality Index (AQI) prediction is widely studied because timely estimates of air quality support environmental monitoring and public-health decision making. Modern machine-learning models can learn nonlinear relationships between pollutant concentrations, meteorological conditions, temporal variables, and AQI. However, a high score on complete observations does not necessarily imply robustness when sensor measurements are missing. In operational monitoring, pollutant observations can be unavailable because of sensor faults, communication interruptions, calibration periods, or incomplete records.

The issue is particularly important for AQI models because the target is strongly related to pollutant concentrations. A model that uses PM2.5, PM10, NO2, SO2, CO, and O3 therefore depends on the availability and quality of these inputs. Previous work on the same NSIT Dwarka dataset showed near-perfect XGBoost performance when pollutant variables were included, while removing them at inference caused a substantial loss in predictive accuracy. Temporal and lagged AQI information retained useful predictive power, but it did not replace the information supplied by current pollutant measurements.

This study extends that analysis from feature dependence to missing-observation robustness. Instead of completely removing pollutants, controlled missingness is introduced at four levels—5%, 10%, 20%, and 30%—and multiple imputation strategies are evaluated under the same XGBoost prediction framework. The objective is to quantify how prediction error changes as pollutant availability decreases and to determine whether model explanations remain stable after imputation.

II. LITERATURE REVIW

Méndez et al. [2] reviewed machine-learning approaches to air-quality forecasting and identified pollutant concentrations and meteorological variables as common predictors. Studies by Pande et al. [3], Ravindiran et al. [4], Natarajan et al. [5], and Suri et al. [6] further demonstrate the application of machine learning and neural networks to AQI prediction in Indian settings. However, prediction accuracy remains the primary focus of these studies, with less emphasis on robustness when pollutant measurements are missing.

Handling missing observations is important in environmental time-series prediction because imputation can affect downstream model performance. Tawabini [7] examined relationships between meteorological conditions and air-quality parameters, while Chen and Guestrin [8] introduced XGBoost, an efficient framework for learning nonlinear relationships in structured data. Lundberg and Lee [9] introduced SHAP for interpreting model predictions through feature attribution.

This study addresses the gap between AQI prediction accuracy and missing-data robustness by imposing 5%, 10%, 20%, and 30% missingness in six pollutant variables. Six imputation methods are evaluated using a common XGBoost model, trained on 2020–2024 data and tested on unseen 2025 observations. SHAP analysis is used to examine how feature contributions change as pollutant availability decreases.

III PROBLEM STATEMENT

AQI prediction models often rely on pollutant observations that may become unavailable because of sensor failures, communication problems, or incomplete monitoring. When one or more pollutant measurements are missing, imputation is required before the data can be supplied to the prediction model. However, different imputation strategies can introduce different levels of error, and the effect of increasing pollutant missingness on AQI prediction performance is not always clear.

A further challenge is that high predictive performance does not necessarily imply that the model's learned feature relationships remain reliable under incomplete observations. In particular, it is important to determine whether the dominant pollutant contributions identified by an explainability method such as SHAP remain stable as the proportion of missing pollutant observations increases.

Therefore, this research investigates the robustness of XGBoost-based AQI prediction under controlled missingness in six pollutant variables: PM2.5, PM10, NO2, SO2, CO, and O3. Missingness levels of 5%, 10%, 20%, and 30% are evaluated using six imputation strategies: Mean, Median,

Forward Fill, Linear Interpolation, KNN, and XGBoost-based imputation. The study further examines whether predictive degradation and feature-attribution stability behave consistently under increasing data incompleteness.

Specifically, this research aims to:

- Quantify AQI prediction degradation using MAE, RMSE, and R^2 under 5–30% missingness.
- Compare six imputation strategies for recovering missing pollutant observations
- Evaluate the robustness of XGBoost-based AQI prediction as pollutant missingness increases.
- Analyze the stability of dominant feature contributions using SHAP.
- Determine whether stable feature attribution corresponds to stable predictive performance under missing pollutant observations.

IV. METHODOLOGY

A. Dataset and preprocessing

The publicly accessible Delhi Pollution AQI dataset [1] provided the AQI data for Delhi. Measurements were taken every six hours (four samples per day) using observations from the NSIT Dwarka monitoring station (2020–2025).

AQI values (0–500), station contains 8,768 observations. The source provides four observations per day at approximately 06:00, 12:00, 18:00, and 23:00; the study retains these observations without resampling to an artificial hourly grid. The target is AQI on the 0–500 scale. Six pollutants—PM2.5, PM10, NO2, SO2, CO, and O3—are treated as potentially missing model inputs. Meteorological variables are temperature, humidity, wind speed, and visibility. Temporal variables include year, month, day, hour, day of week, weekend status, and season.

The records are sorted chronologically. Historical AQI features are generated using row shifts corresponding to approximately 6-hour, 12-hour, 24-hour, 48-hour, and one-week history: aqi_lag_1, aqi_lag_2, aqi_lag_4, aqi_lag_8, and aqi_lag_28. The first 28 observations are removed because the weekly lag cannot be computed, producing 8,740 modeling observations. Day-of-week is encoded numerically from Monday=0 to Sunday=6, and season is represented by dummy variables.

Component	Details
Dataset	Delhi Pollution AQI Dataset
Source	Mehta, 2022 [1]
Region	Delhi NCR
Selected Station	NSIT Dwarka
Target Variable	AQI
Data Type	six-hourly observations (4 observations/day)
Pollutant Features	PM2.5, PM10, NO2, SO2, CO, O3
Meteorological Features	Temperature, Humidity, Wind Speed, Visibility
Temporal Features	Hour, Day, Month, Year, Day of Week, Season
Historical Features	Lagged AQI observations and other historical/lag-based features

Table 1. Dataset Components

Item	Configuration
Station	NSIT Dwarka, Delhi
Original observations	8,768
Modeling observations	8,740
Training period	2020–2024
Test period	2025
Missingness levels	5%, 10%, 20%, 30%
Missing variables	PM2.5, PM10, NO2, SO2, CO, O3
Prediction model	XGBoost, 300 estimators, depth 6

Table 2. Dataset and experimental configuration.

B. Missingness and imputation design

Missingness is introduced only into the six pollutant variables. The AQI target, temporal variables, meteorological variables, and lagged AQI features remain complete. Four missingness levels are evaluated: 5%, 10%, 20%, and 30%. A fixed random seed is used so that the experiments are reproducible and comparable across methods. For each missingness level, the same incomplete dataset is processed independently by six imputation methods.

- Mean: replaces missing values using the mean of the corresponding pollutant.
- Median: replaces missing values using the corresponding pollutant median.
- Forward Fill: propagates the most recent available observation through missing positions.
- Linear Interpolation: estimates missing values from neighboring observations.
- KNN: uses five nearest observations to estimate missing pollutant values.
- XGBoost imputation: trains a separate XGBoost regressor for each pollutant using the remaining available variables.

C. Prediction model and temporal evaluation

XGBoost is used as the single prediction model so that differences in results primarily reflect missingness and imputation rather than differences between model architectures. The model uses pollutant, meteorological, temporal, seasonal, and lagged-AQI features. Datetime and date strings are excluded from the predictor matrix because their information is already represented by numerical temporal variables. The XGBoost configuration uses 300 estimators, maximum depth 6, learning rate 0.05, subsampling 0.8, column subsampling 0.8, squared-error objective, and random state 42.

To prevent random train–test mixing across time, all observations from 2020–2024 are used for training and the unseen 2025 observations are used for testing. This preserves the chronological structure of the experiment and provides a consistent test period for all imputation levels. A complete-data model is trained first as the 0% missingness reference.

D. Evaluation and explainability

Performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2). MAE provides a direct measure of average absolute prediction error, RMSE emphasizes larger errors, and R^2 measures the proportion of target variance explained by the model. Relative degradation is calculated with respect to the

complete-data MAE using ((MAE_m−MAE_0)/MAE_0)×100. To summarize resilience across the four missingness levels, the average MAE is calculated for each imputation method and transformed into a relative 0–100 resilience scale using the observed maximum and minimum average MAE among the six methods.

SHAP TreeExplainer is applied to the trained XGBoost models. Mean absolute SHAP values are used to quantify global feature importance on the 2025 test set. SHAP analysis is performed for the complete-data baseline and Mean-imputed datasets at 5%, 10%, 20%, and 30% missingness. This focused analysis tests whether the dominant feature structure changes as pollutant observations become increasingly incomplete.

V. RESULTS AND DISCUSSION

A. Prediction performance under missing pollutant observations

Table 1 summarizes the complete-data baseline and all 24 missingness–imputation experiments. The complete model achieves MAE 0.7774, RMSE 1.2228, and R^2 0.9999. Introducing missing pollutant observations causes a progressive increase in prediction error for every imputation strategy. The effect is already visible at 5% missingness and becomes substantially larger at 20% and 30%.

Missingness	Imputation	MAE	RMSE	R^2
0%	Complete	0.7774	1.2228	0.9999
5%	Mean	1.8709	4.744	0.9992
5%	Median	1.8329	4.8138	0.9992
5%	Forward Fill	2.9429	7.0263	0.9983
5%	Linear Interpolation	2.6895	6.9966	0.9983
5%	KNN	1.9901	5.2001	0.999
5%	XGBoost	2.1545	6.1768	0.9987
10%	Mean	2.5644	5.1934	0.9991
10%	Median	2.5513	5.4533	0.999
10%	Forward Fill	4.6489	9.9922	0.9965
10%	Linear Interpolation	4.2081	9.2483	0.997
10%	KNN	3.4148	7.8862	0.9978
10%	XGBoost	3.076	7.7301	0.9979
20%	Mean	4.7672	9.9115	0.9965
20%	Median	4.8209	10.0332	0.9965
20%	Forward Fill	8.1357	15.1655	0.9919
20%	Linear Interpolation	7.2784	14.0436	0.9931
20%	KNN	6.7077	14.6386	0.9925
20%	XGBoost	5.2015	11.2365	0.9956
30%	Mean	6.1867	11.871	0.995
30%	Median	6.3623	11.8705	0.995
30%	Forward Fill	10.1538	16.677	0.9902
30%	Linear Interpolation	9.5282	15.6055	0.9914
30%	KNN	8.5779	15.879	0.9911
30%	XGBoost	6.9432	13.4388	0.9936

Table 3. XGBoost performance on the 2025 test set under controlled pollutant missingness.

At 5% missingness, Mean and Median imputation produce MAE values of 1.8709 and 1.8329, respectively, while Forward Fill and Linear Interpolation produce larger errors of 2.9429 and 2.6895. At 30% missingness, the corresponding MAEs rise to 6.1867, 6.3623, 10.1538, and 9.5282. KNN and XGBoost-based imputation occupy an intermediate range.

Thus, the ranking of methods is not determined only by whether an imputer is model-based; simple distribution-based replacement remains highly competitive for this dataset.

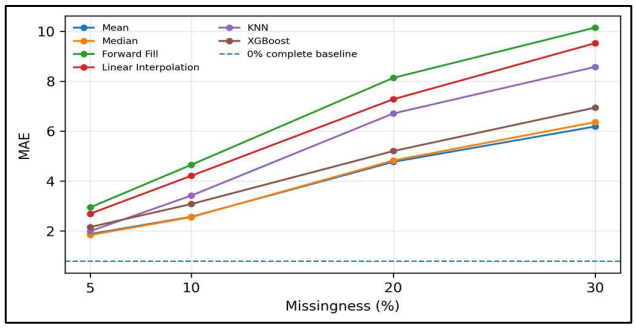


Fig. 1. MAE variation with increasing pollutant missingness

B. Robustness and relative degradation

Figure 1 shows a monotonic increase in MAE for every imputation method. The complete-data baseline is substantially lower than all missing-data cases, confirming that imputation cannot fully reproduce the information available from measured pollutant concentrations. The relative degradation from the complete-data baseline at 30% missingness is 695.8% for Mean, 718.5% for Median, 1,206.9% for Forward Fill, 1,125.4% for Linear Interpolation, 1,003.2% for KNN, and 793.0% for XGBoost imputation.

Although these percentages are large because the complete-data MAE is very small, the absolute errors remain within the AQI scale and R^2 remains above 0.99 for all methods at 30% missingness. This distinction is important: relative degradation highlights sensitivity to the highly accurate baseline, whereas MAE and R^2 describe the resulting prediction quality directly.

Imputation Method	Average MAE	Relative Resilience
Mean	3.8473	100.0
Median	3.8919	98.3
XGBoost	4.3438	81.07
KNN	5.1726	49.47
Linear Interpolation	5.9261	20.75
Forward Fill	6.4703	0.0

Table 4. Average MAE and relative resilience across 5–30% missingness.

Table 3 provides a compact summary of robustness across all missingness levels. Mean imputation has the lowest average MAE (3.8473) and therefore receives the maximum relative resilience value on the six-method scale. Median is very close (3.8919), followed by XGBoost imputation (4.3438), KNN (5.1726), Linear Interpolation (5.9261), and Forward Fill (6.4703). The resilience score is relative to the six tested methods and is not an absolute quality score.

VI. SHAP-BASED EXPLAINABILITY

SHapley Additive exPlanations (SHAP) were employed to investigate how pollutant missingness affects the contribution of individual features to AQI predictions. The analysis was conducted at five missingness levels (0%, 5%, 10%, 20%, and 30%) using mean imputation. Mean absolute SHAP values were used to quantify the relative influence of each feature on the model predictions, while normalized importance percentages facilitated comparisons across missingness levels. The results are summarized in Fig. 2 and Table 5.

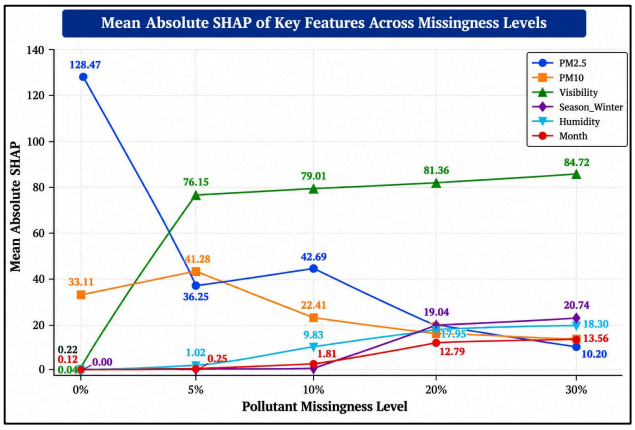


Fig. 2. Mean Absolute SHAP Values of Key Features Across Pollutant Missingness Levels

Under complete-data conditions, PM2. 5 was the dominant predictor, with a mean absolute SHAP value of 128.47 and an importance of 77.40%. PM10 was the second most influential feature, contributing 19.95%, while SO2 accounted for approximately 1.32%. Together, PM2.5 and PM10 represented approximately 97.35% of the total absolute SHAP importance. This indicates that the model's predictions under complete-data conditions were primarily driven by particulate matter concentrations.

Table 5. Absolute SHAP Values and Feature Importance Across Pollutant Missingness Levels

Feature	0%		5%		10%		20%		30%	
	SHAP	Imp. %	SHAP	Imp. %	SHAP	Imp. %	SHAP	Imp. %	SHAP	Imp. %
pm25	**128.470093**	**77.398178**	**36.250847**	**21.204611**	**42.691517**	**23.295361**	**19.039749**	**9.618023**	**10.203402**	**5.175414**
pm10	**33.109936**	**19.947435**	**41.283672**	**24.148516**	**22.414213**	**12.230701**	**15.869041**	**8.016324**	**13.563265**	**6.879618**
so2	2.192983	1.321186	4.364701	2.553093	4.472926	2.440729	2.834658	1.431942	2.863868	1.452624
no2	0.790407	0.476189	5.510538	3.223340	3.838522	2.094555	3.626226	1.831806	5.213224	2.644274
co	0.529424	0.318957	3.833253	2.242227	8.143135	4.443442	3.469225	1.752496	3.298812	1.673238
visibility	**0.223125**	**0.134424**	**76.154800**	**44.546070**	**79.005028**	**43.110451**	**81.364441**	**41.101646**	**84.720764**	**42.972435**
season_summer	0.189355	0.114079	1.195571	0.699338	6.585584	3.593537	11.213235	5.664421	11.361092	5.762622
month	**0.123854**	**0.074617**	**0.261363**	**0.152882**	**1.807164**	**0.986110**	**12.793572**	**6.462736**	**13.353701**	**6.773322**
o3	0.092372	0.055650	0.196664	0.115037	0.333388	0.181919	0.778778	0.393404	0.587824	0.298158
temperature	0.089029	0.053637	0.210594	0.123185	2.899994	1.582431	6.195384	3.129628	8.247114	4.183137
year	0.048778	0.029387	0.057021	0.033354	0.079362	0.043305	0.181761	0.091817	0.471645	0.239230
humidity	**0.036800**	**0.022171**	**1.019937**	**0.596603**	**9.828675**	**5.363186**	**17.948948**	**9.066999**	**18.297913**	**9.281147**
wind_speed	0.025166	0.015162	0.146788	0.085863	0.256955	0.140212	0.302585	0.152852	0.358668	0.181925
day	0.024572	0.014803	0.089963	0.052623	0.176510	0.096316	0.698473	0.352837	0.945474	0.479567
day_of_week	0.022777	0.013722	0.065128	0.038096	0.088300	0.045454	0.307639	0.155406	0.430126	0.218171
hour	0.014389	0.008669	0.075115	0.043938	0.197321	0.107671	0.304927	0.154035	0.758338	0.384648
season_winter	**0.001183**	**0.000713**	**0.209463**	**0.122523**	**0.416621**	**0.227336**	**19.528620**	**9.864979**	**20.741453**	**10.520569**
season_post_monsoon	0.000978	0.000589	0.020786	0.012159	0.011825	0.006452	1.468300	0.741719	1.620379	0.821896
is_weekend	0.000714	0.000430	0.011179	0.006539	0.019856	0.010835	0.033507	0.016926	0.114355	0.058004

As pollutant missingness increased, a substantial shift in feature importance was observed. At 5% missingness, visibility became the most influential feature, accounting for 44.55% of total importance, while PM2.5 and PM10 contributed 21.20% and 24.15%, respectively. At 10% missingness, visibility retained its dominant role with an importance of 43.11%, whereas PM2.5 and PM10 contributed 23.30% and 12.23%, respectively. This pattern persisted at higher missingness levels: visibility accounted for 41.10% and 42.97% of total importance at 20% and 30% missingness, respectively. Its mean absolute SHAP value increased from 0.22 under complete-data conditions to 84.72 at 30% missingness, indicating a marked change in the model's reliance on this feature.

Meteorological and seasonal features also gained importance as pollutant observations became increasingly unavailable. Humidity increased from 0.02% importance at baseline to 9.28% at 30% missingness. Similarly, the importance of the winter-season indicator rose from approximately 0.001% to 10.52%, while month increased from 0.07% to 6.77%. These changes suggest that the model increasingly relied on visibility, humidity, and seasonal information when pollutant measurements were missing. In contrast, the importance of PM2.5 declined to 5.18% at 30% missingness, while PM10 importance decreased to 6.88%.

Overall, the SHAP analysis reveals a pronounced change in the model's predictive feature dependence under increasing pollutant missingness. Although particulate matter dominates predictions when observations are complete, visibility and meteorological or seasonal features become more influential under missing-data conditions. These findings highlight the importance of examining model explanations alongside predictive performance when evaluating AQI forecasting under incomplete observations. However, SHAP importance represents model attribution rather than causal influence. The unusually large increase in visibility importance should therefore be interpreted cautiously and investigated further to determine whether it reflects genuine predictive information, correlations among features, or changes in the model's learned behavior under imputation.

VII. DISCUSSION

Three key findings emerge from the controlled experiments. First, increasing pollutant missingness generally degrades predictive performance because the XGBoost model relies directly on pollutant concentrations to estimate AQI. As the proportion of missing observations increases, imputation becomes increasingly important for preserving the information required for accurate prediction. Although imputed values cannot perfectly reproduce every original observation, the choice of imputation strategy can help mitigate the resulting loss of predictive accuracy.

Second, the performance of imputation methods varies with the missingness level. Mean imputation was selected for the SHAP-based explainability analysis because it achieved the best average MAE across the evaluated experimental conditions. Median imputation also demonstrated competitive performance, whereas Forward Fill and Linear Interpolation exhibited greater error growth as missingness increased. Selecting Mean imputation therefore provides a consistent basis for examining how feature contributions change under incomplete pollutant observations while keeping the explainability analysis focused on the strongest overall-performing imputation strategy.

Third, the SHAP analysis provides further insight into the model's predictive behavior under missingness. Under complete-data conditions, PM2.5 and PM10 dominate the explanations, contributing 77.40% and 19.95% of total feature importance, respectively. As missingness increases under Mean imputation, visibility becomes the dominant feature, accounting for approximately 41–45% of total importance across the 5–30% missingness levels. Humidity, the winter-season indicator, and month also become more influential at higher missingness levels, while the relative contributions of PM2.5 and PM10 decline. These results indicate that the model adapts its predictive reliance across available features when pollutant observations are incomplete. The combination of predictive-performance evaluation and SHAP analysis consequently provides complementary evidence: MAE quantifies prediction accuracy, while SHAP characterizes the relative contributions of the predictors used by the model.

The study has several limitations. Missingness is synthetically imposed and may not capture every mechanism responsible for real-world sensor outages. The experiments evaluate a finite set of global missingness levels and use the

same XGBoost architecture across conditions. Furthermore, the explainability analysis focuses on Mean imputation, selected on the basis of its best average MAE, rather than comparing SHAP attributions across every imputation method. Future work could extend the analysis to structured and sensor-specific missingness, uncertainty-aware imputation, and additional model architectures while preserving the chronological evaluation protocol.

VII. PROPOSED FUTURE DEPLOYMENT

The findings of this study provide a foundation for developing a robust AQI prediction system capable of operating under incomplete pollutant observations. The experiments demonstrate that XGBoost can maintain an (R^2) above 0.99 even at 30% pollutant missingness when an appropriate imputation strategy is applied, although prediction errors increase as data availability decreases. Mean imputation achieved the lowest average MAE of 3.8473 across the evaluated missingness levels, making it the preferred candidate for an initial deployment framework.

Building on these results, future deployment will integrate the selected imputation strategy with the trained XGBoost prediction model. Incoming observations from air-quality monitoring stations will first undergo data validation to identify missing or invalid pollutant measurements. Missing values will then be handled using the mean parameters estimated from the training data, after which the processed features will be supplied to the prediction model to generate AQI estimates. The same feature definitions and preprocessing procedures used during training will be maintained during inference to ensure consistency.

The experimental results also establish the importance of monitoring model behavior, rather than evaluating prediction accuracy alone. Accordingly, the proposed system will record the proportion of missing pollutant observations and track prediction errors whenever reference AQI values become available. SHAP-based analysis will be used periodically to examine changes in feature contributions under incomplete-data conditions. In particular, the observed increase in the importance of visibility and meteorological or seasonal variables under Mean imputation provides a reason to monitor whether similar attribution patterns occur during operational use.

The proposed deployment workflow is therefore:

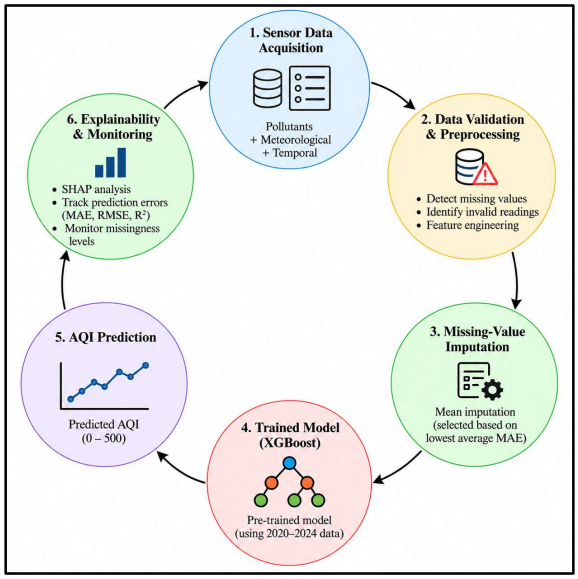


Fig. 3. Proposed Workflow for AQI Prediction Under Missing Pollutant Observations

Future validation will extend the current controlled experiments to real sensor outages and changing environmental conditions. If sustained prediction degradation is observed, alternative imputation strategies or model updates can be evaluated using validated observations before changes are introduced into the operational pipeline. This approach translates the study's findings into a practical deployment direction while retaining the study's emphasis on prediction accuracy, missing-data robustness, and model interpretability.

VII. COMPARISON WITH PRIOR WORK

Existing studies have primarily focused on improving AQI prediction accuracy through machine-learning, deep-learning, and optimized predictive models. Pande et al. [2] evaluated machine-learning and deep-learning techniques for daily AQI prediction in Delhi, while Ravindiran et al. [3] investigated machine-learning-based air-quality prediction in Visakhapatnam. Natarajan et al. [4] explored optimized

machine-learning models for AQI prediction across major Indian cities, and Suri et al. [5] examined a neural-network-based approach to air-quality prediction. Tawabini [6] studied relationships between meteorological conditions and air-quality parameters. These studies provide a foundation for applying predictive models to air-quality analysis, while Chen and Guestrin [7] introduced XGBoost, a scalable gradient-boosting method, and Lundberg and Lee [8] proposed SHAP's unified framework for interpreting model predictions.

In contrast to studies primarily concerned with predictive performance or relationships between environmental variables, the present study investigates the robustness and interpretability of AQI prediction under artificially introduced pollutant missingness. Six imputation methods—mean, median, forward fill, linear interpolation, K-nearest neighbors (KNN), and XGBoost-based imputation—are evaluated at missingness levels of 5%, 10%, 20%, and 30%. Mean imputation achieves the lowest average MAE of 3.8473, followed by median imputation at 3.8919, XGBoost imputation at 4.3438, KNN at 5.1726, linear interpolation at 5.9261, and forward fill at 6.4703. Furthermore, SHAP analysis is used to investigate how missing pollutant observations and their imputation alter the relative importance attributed to pollutant, meteorological, and temporal features. This combined evaluation provides insight into both predictive accuracy and changes in model interpretation under incomplete pollutant observations.

VIII. CONCLUSION

This study investigated the impact of missing pollutant observations on machine-learning-based AQI prediction using six imputation methods at missingness levels of 5%, 10%, 20%, and 30%. An XGBoost model was evaluated using temporally separated training and testing data, with 2020–2024 used for training and 2025 reserved for testing. The complete-data baseline achieved an MAE of 0.7774, an RMSE of 1.2228, and an \(R^2\) of 0.9999. Among the six imputation methods, mean imputation achieved the lowest average MAE of 3.8473, followed by median imputation (3.8919) and XGBoost-based imputation (4.3438). Forward fill produced the highest average MAE of 6.4703, indicating that its performance was comparatively weaker under the evaluated missingness conditions.

The SHAP analysis further demonstrated that missing pollutant observations and the applied imputation strategy can substantially alter the distribution of feature importance. While PM2.5 and PM10 dominated the complete-data baseline explanations, visibility became the most influential feature in the mean-imputation experiments, accounting for over 40% of normalized importance across the tested missingness levels. This shift highlights that maintaining predictive performance alone may not ensure consistency in model interpretation when pollutant data are incomplete.

Overall, the findings indicate that mean imputation was the most effective of the six evaluated methods in terms of average MAE for this dataset and experimental setup. The study also demonstrates the importance of examining feature-attribution changes alongside conventional prediction metrics when assessing AQI models under missing data. Future work should evaluate additional imputation techniques, test the framework on independent monitoring stations and datasets, and investigate alternative missingness mechanisms to assess the generalizability of these findings.

REFERENCES

[1] V. Mehta, "Delhi Pollution AQI Dataset," Kaggle, 2020–2025. https://www.kaggle.com/datasets/vishardmehta/delhi-pollution-aqi-dataset

[2] C. B. Pande, L. Radhadevi, and M. B. Satyanarayana, "Evaluation of machine learning and deep learning models for daily air quality index prediction in Delhi city, India," *Environmental Monitoring and Assessment*, vol. 196, no. 12, Art. no. 1215, 2024, doi: 10.1007/s10661-024-13351-1.

[3] G. Ravindiran, G. Hayder, K. Kanagarathinam, A. Alagumalai, and C. Sonne, "Air quality prediction by machine learning models: A predictive study on the Indian coastal city of Visakhapatnam," *Chemosphere*, vol. 338, Art. no. 139518, 2023, doi: 10.1016/j.chemosphere.2023.139518.

[4] S. K. Natarajan, P. Shanmurthy, D. Arockiam, B. Balusamy, and S. Selvarajan, "Optimized machine learning model for air quality index prediction in major cities in India," *Scientific Reports*, vol. 14, Art. no. 6795, 2024, doi: 10.1038/s41598-024-54807-1.

[5] R. S. Suri, A. K. Jain, N. R. Kapoor, A. Kumar, H. C. Arora, K. Kumar, and H. Jahangir, "Air quality prediction—A study using neural network based approach," *Journal of Soft Computing in Civil Engineering*, vol. 7, no. 1, pp. 93–113, 2023, doi: 10.22115/SCCE.2022.352017.1488.

[6] B. Tawabini, "Using machine learning algorithms to study the relationship between meteorological conditions and air quality parameters," *Scientific Reports*, vol. 16, Art. no. 10392, 2026, doi: 10.1038/s41598-026-39579-0.

[7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[8] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 4765–4774.